Compliance sheet — star-cw

Everything a supplier qualification, a legal review or an export check usually asks about star-cw 0.1.0, in one document.

1. Identity

Document	STAR-PKG-STAR-CW-EN · v1.3 · 2026-10-16
Package	star-cw
Version	0.1.0
Capability	star.orbit.relative-cw
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_cw

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_cw-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_cw-0.1.0-py3-none-any.whl	d95578b636d0c379dcd7ed4f69ce1b9ff6b7cba3c0f61f873ec34f18c2afe0f0
star_cw.py	f555f308539d79c45c15d29a258b0dde9f5ced0869ef4c3542706611a10e2999
star_cw-0.1.0-source.zip	c6b1f0070946e9f604c765761701be4ce56659e0559da57addc8a9d8a097717a

5. What it does, and its limits

Closed-form Clohessy-Wiltshire transition matrix, free drift and two-impulse rendezvous in the chief's rotating frame; algebra equal to the matrix exponential within 1.1e-13, departure from real two-body motion measured per separation, singular transfer times refused.

Circular chief only (no Tschauner-Hempel), no J2, drag or thrust arcs, no collision check along the transfer, no optimisation of the transfer time. A found defect is recorded in the module: the first version returned an empty tuple from `propagate`; the cross-check now refuses answers that do not have six components.

- Frame and state: x radial outward, y along the velocity, z along the orbit normal, origin at the chief; state (x, y, z, vx, vy, vz) relative to the rotating frame; n in rad/s; any consistent length unit.
- `stm(n, t)` (6×6) and `propagate(state, n, t)`: the closed-form solution; agrees with SciPy's matrix exponential of the system matrix within 1.1e-13 relative on 300 states and times up to three revolutions.
- `rendezvous(r0, v0, rf, n, t)` returns (dv1, dv2): the impulse now that brings the deputy to the point rf of the rotating frame after t, and the impulse there that stops it relative to rf.
- Every function returns finite floats of the documented shape or raises `ValueError`: non-numeric, boolean, NaN, infinite or |value| >= 1e150 input; wrong shapes; n <= 0; t <= 0 for a rendezvous; and the singular transfer times (whole and half revolutions, and the other roots of 8(1 - cos nt) = 3 nt sin nt), where no finite impulse reaches an arbitrary point.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_cw_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-cw}: Relative motion near a circular orbit (Clohessy-Wiltshire)},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_cw/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

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